

For questions 1-6, factor each expression by grouping.

1. $2y^4 - 8y^3 + 10y^2 - 40y$

$$\begin{aligned} &2y^3(y - 4) + 10y(y - 4) \\ &(2y^3 + 10y)(y - 4) \\ &2y(y^2 + 5)(y - 4) \end{aligned}$$

2. $15x^2 + 62x + 63$

$$\begin{aligned} &15x^2 + 35x + 27x + 63 \\ &5x(3x + 7) + 9(3x + 7) \\ &(5x + 9)(3x + 7) \end{aligned}$$

3. $-12a + 20ab - 35b + 21$

$$\begin{aligned} &4a(-3 + 5b) - 7(5b - 3) \\ &(4a - 7)(5b - 3) \end{aligned}$$

4. $6x^2 - x - 77$

$$\begin{aligned} &6x^2 - 22x + 21x - 77 \\ &2x(3x - 11) + 7(3x - 11) \\ &(2x + 7)(3x - 11) \end{aligned}$$

5. $3x^2 + 12x + 6x + 24$

$$\begin{aligned} &3x(x + 4) + 6(x + 4) \\ &(3x + 6)(x + 4) \\ &3(x + 2)(x + 4) \end{aligned}$$

6. $4x^2 + 38x + 18$

$$\begin{aligned} &2(2x^2 + 19x + 9) \\ &2(2x^2 + 18x + x + 9) \\ &2(2x(x + 9) + 1(x + 9)) \\ &2(2x + 1)(x + 9) \end{aligned}$$

For questions 7-10, consider the trinomial $15x^2 - 34x - 16$.

7. List all the factor pairs for -240 .

$-1, 240$	$-6, 40$	$1, -240$	$6, -40$
$-2, 120$	$-8, 30$	$2, -120$	$8, -30$
$-3, 80$	$-10, 24$	$3, -80$	$10, -24$
$-4, 60$	$-12, 20$	$4, -60$	$12, -20$
$-5, 48$	$-15, 16$	$5, -48$	$15, -16$

8. Find the factor pair for -240 that add to the sum of -34 .

$-40, 6$

9. Use question 8 to write the expanded form of $15x^2 - 34x - 16$ that can be used to factor the expression by grouping.

$15x^2 - 40x + 6x - 16$

10. Factor the expression $15x^2 - 34x - 16$ by grouping.

$15x^2 - 40x + 6x - 16$
 $5x(3x - 8) + 2(3x - 8)$
 $(5x + 2)(3x - 8)$